

DBMS Module 3 Complete Normalization Notes

C.V. Raman Global University • Computer Science & Engineering • Semester 3

1. Fundamentals of Database Normalization

Database normalization is the systematic approach of decomposing tables to eliminate data redundancy and undesirable anomalies (Insertion, Deletion, and Update anomalies).

Key Concepts:

- Prime Attribute: Attribute that is part of any candidate key.
- Non-Prime Attribute: Attribute not belonging to any candidate key.

2. Normal Form Hierarchy Summary

- 1NF: Atomic domain values, no multivalued or repeating groups.
- 2NF: 1NF + No partial dependency (No non-prime attribute depends on a proper subset of any candidate key).
- 3NF: 2NF + No transitive dependency (For every $X \rightarrow Y$, either X is a superkey or Y is a prime attribute).
- BCNF: For every functional dependency $X \rightarrow Y$, X must be a superkey.